

Statistical Learning Techniques for Credit Loan Approval: A Comprehensive Analysis and Predictive Modeling Approach

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January 27, 2025

1 Introduction

The goal of this project is to conduct a comprehensive analysis of a credit loan approval dataset, with the dual purpose of understanding the underlying structure of the data and predicting loan approval outcomes. The dataset includes a variety of features related to loan applicants, such as demographic characteristics, financial information, and credit history, which are used to determine whether an application is approved or rejected. By applying both exploratory and predictive modeling techniques, we aim to extract meaningful insights and build accurate prediction models.

The analysis is divided into two main phases: **unsupervised learning** and **supervised learning**.

In the first phase, we focus on understanding the structure of the data using unsupervised techniques:

- We begin by checking for **multicollinearity** to identify and remove highly correlated features, which can distort the performance of models.
- **Principal Component Analysis (PCA)** is applied to reduce the dimensionality of the data while retaining as much of the variance as possible.
- **Hierarchical clustering** is performed to detect any natural groupings in the data, providing insights into hidden patterns.
- **t-SNE** and **UMAP** are employed to visualize high-dimensional data in two-dimensional spaces, helping us better understand the clustering of applicants.

In the second phase, we build and evaluate predictive models using **supervised learning** techniques:

- **Logistic regression** with Elastic Net regularization is used to identify significant features and create a robust, interpretable model.
- A **Random Forest** model is trained to capture complex, non-linear relationships and to evaluate the importance of different variables.
- **XGBoost** is applied for its ability to handle high-dimensional data and its effectiveness in improving model performance through boosting techniques.
- Finally, a **neural network** is trained with cross-validation to explore the performance of deep learning models for loan approval prediction.

The overarching aim of this analysis is twofold: to predict loan approval decisions with high accuracy and to uncover the most influential factors driving these decisions. By comparing the performance of different models, we will determine which techniques offer the best balance of interpretability and predictive power.

2 Dataset Description

The dataset used in this project is sourced from Kaggle, specifically from the following repository: Credit Card Approval Clean Data by Samuel Cortinhas. This dataset contains information on credit card applications, including various applicant attributes and the corresponding approval status. The original dataset consists of 16 columns and 690 rows.

2.1 Original Dataset Structure

The original dataset includes the following features:

- **Gender (Categorical):** Applicant's gender, encoded as a binary integer (1 for Male, 0 for Female).
- **Age (Continuous):** Applicant's age.
- **Debt (Continuous):** Total amount of debt the applicant holds.
- **Married (Categorical):** Whether the applicant is married, encoded as a binary integer (1 for Yes, 0 for No).
- **BankCustomer (Categorical):** Whether the applicant is a customer of the bank, encoded as a binary integer (1 for Yes, 0 for No).
- **Industry (Categorical):** The industry in which the applicant works, represented as a categorical string (e.g., "Industrials", "Materials").
- **Ethnicity (Categorical):** The applicant's ethnicity, represented as a categorical string (e.g., "White", "Black").
- **YearsEmployed (Continuous):** Number of years the applicant has been employed.
- **PriorDefault (Categorical):** Whether the applicant has previously defaulted on a loan, encoded as a binary integer (1 for Yes, 0 for No).
- **Employed (Categorical):** Whether the applicant is currently employed, encoded as a binary integer (1 for Yes, 0 for No).
- **CreditScore (Continuous):** The applicant's credit score, represented as an integer value.
- **DriversLicense (Categorical):** Whether the applicant holds a valid driver's license, encoded as a binary integer (1 for Yes, 0 for No).
- **Citizen (Categorical):** The citizenship status of the applicant, represented as a categorical string (e.g., "ByBirth", "ByOtherMeans").
- **ZipCode (Categorical):** The applicant's zip code.

- **Income (Continuous):** Annual income of the applicant.
- **Approved (Categorical):** Whether the loan application was approved, encoded as a binary integer (1 for Approved, 0 for Rejected).

2.2 Data Preprocessing Pipeline

In order to prepare the dataset for further analysis, the following steps were undertaken to clean and process the data:

2.2.1 Conversion of Categorical Variables into Factors

Categorical variables in the dataset, such as *Industry*, *Ethnicity*, *Citizen*, *Gender*, and others, were converted into factor variables to be properly handled in models. This step ensures that these categorical variables are treated as discrete categories rather than continuous data. The specific columns transformed into factors include:

- **Industry:** Industry type the applicant works in.
- **Ethnicity:** The ethnicity group of the applicant.
- **Citizen:** Citizenship status of the applicant.
- **Approved:** Loan approval status (0: Rejected, 1: Approved).
- **Gender:** Gender of the applicant.
- **Married:** Marital status of the applicant.
- **BankCustomer:** Whether the applicant is a bank customer.
- **PriorDefault:** Whether the applicant has a prior default record.
- **Employed:** Employment status of the applicant.
- **DriversLicense:** Whether the applicant has a driver's license.

The structure of the dataset was verified using the `str()` function after converting these variables into factors.

2.2.2 Identification of Numeric Features for Scaling

To ensure that features with larger numerical ranges do not dominate models that rely on distance-based metrics, numeric features were identified and scaled. The scaling process centered and scaled each feature (i.e., standardization), making them more suitable for machine learning algorithms. Features such as *Age*, *Debt*, *YearsEmployed*, and *CreditScore* were included in this step, while categorical variables such as *Approved* were excluded.

2.2.3 Scaling of Numeric Features

A preprocessing function, `preProcess()`, was applied to center and scale all numeric features. This ensured that the numerical values were normalized, with a mean of 0 and a standard deviation of 1, improving model convergence during training.

2.2.4 One-hot Encoding of Categorical Variables

Certain categorical variables, specifically *Industry*, *Ethnicity*, and *Citizen*, were one-hot encoded to transform them into a binary format suitable for machine learning algorithms. The `dummyVars()` function was used for this purpose. One-hot encoding allows the model to interpret the categorical variables without imposing any ordinal relationships between categories.

2.3 Multicollinearity Check

During the formal multicollinearity analysis, we identified that the *BankCustomer* and *Married* columns were highly correlated with other features in the dataset. These two columns were removed to reduce redundancy and prevent multicollinearity from affecting the model's performance. The decision to remove these variables was made after running the initial multicollinearity check, as they exhibited high correlation with other features. This step helped streamline the subsequent analysis.

After removing *BankCustomer* and *Married*, we proceeded with a detailed multicollinearity check, which involved analyzing the correlation matrix and calculating the Variance Inflation Factor (VIF).

2.3.1 Correlation Matrix

To detect any remaining multicollinearity between the numeric variables, I calculated a correlation matrix for the dataset. A heatmap was generated to visualize the relationships between the numeric features, helping to identify any further potential issues with multicollinearity:

In the heatmap, correlations closer to -1 or 1 indicate strong relationships between variables. Based on this analysis, variables with correlation coefficients exceeding 0.7 were flagged for removal to avoid multicollinearity.

2.3.2 Identifying Highly Correlated Variables

Following the correlation matrix analysis, I identified additional features that were highly correlated (correlation coefficient > 0.7). These variables were:

- *Industry.Healthcare*
- *Citizen.ByBirth*

These features were removed from the dataset to further mitigate multicollinearity issues.

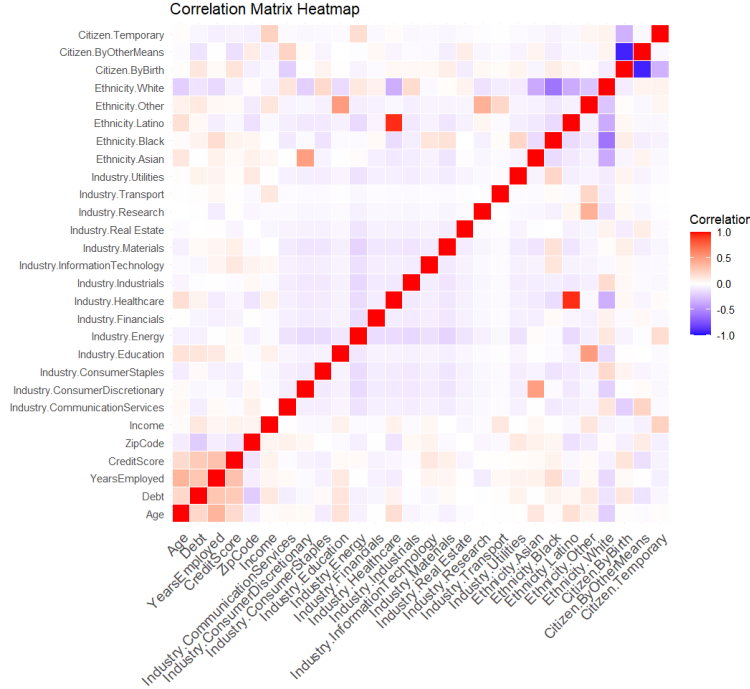


Figure 1: Correlation Matrix Heatmap for Numeric Variables

2.3.3 Refitting the Logistic Regression Model and Calculating VIF

After removing the aliased variables (*Industry.Utilities*, *Ethnicity.White*, and *Citizen.Temporary*), the logistic regression model was refitted using the cleaned dataset. To further assess multicollinearity, I calculated the Variance Inflation Factor (VIF) for each remaining variable in the model.

2.3.4 Plotting VIF Values

The VIF values were visualized to identify any variables with high multicollinearity ($VIF > 5$). A bar plot was generated, with a dashed red line indicating the VIF threshold of 5:

Since *Industry.Energy* had a VIF value exceeding the threshold, it was removed to improve the model's reliability. This step ensured that multicollinearity was minimized, allowing for more stable and interpretable model coefficients.

2.3.5 Final Model and Dataset

After the removal of *Industry.Energy*, the logistic regression model was refitted once again. The final version of the model no longer exhibited any multicollinearity issues, as confirmed by the VIF values. The cleaned and processed

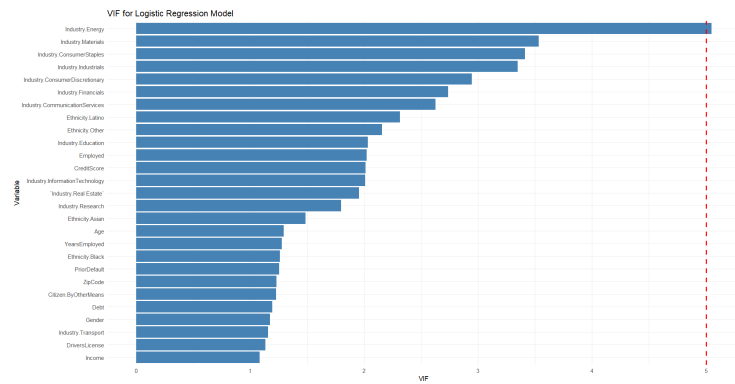


Figure 2: VIF for Logistic Regression Model After Removing Aliased Variables

dataset, now free from multicollinearity, was used for further analysis, providing reliable insights into the factors affecting loan approval.

3 Unsupervised Analysis

In this section, we apply several unsupervised learning techniques to explore the underlying structure of the credit loan dataset. These methods allow us to detect patterns, reduce dimensionality, and identify potential clusters in the data without relying on predefined labels. We use techniques such as PCA, FAMD, hierarchical clustering, t-SNE, and UMAP to visualize and better understand the relationships between features. This analysis provides insights into the data's structure and prepares us for more informed supervised modeling.

3.1 Variable Selection for Unsupervised Analysis

In order to simplify the analysis and focus on the most interpretable and significant features, a subset of the dataset was selected based on the variables that provide critical insights into credit loan approval decisions. These features were chosen for their relevance to both the applicant's financial stability and past behavior, as well as their influence on the approval outcome. The selected variables for unsupervised analysis are as follows:

- **Gender:** Gender of the applicant.
- **Age:** Age of the applicant (continuous numeric variable).
- **Debt:** Total debt of the applicant (continuous numeric variable).
- **YearsEmployed:** Number of years the applicant has been employed (continuous numeric variable).
- **PriorDefault:** Indicates whether the applicant has defaulted previously (binary variable: 1 for Yes, 0 for No).
- **Employed:** Indicates whether the applicant is currently employed (binary variable: 1 for Yes, 0 for No).
- **CreditScore:** Credit score of the applicant (continuous numeric variable).
- **Income:** Annual income of the applicant (continuous numeric variable).
- **Citizen.ByOtherMeans:** Whether the applicant is a citizen by other means (binary variable: 1 for Yes, 0 for No).
- **Approved:** Whether the application was approved (binary variable: 1 for Approved, 0 for Rejected).

By focusing on these key features, we aim to provide a clearer interpretation of the underlying patterns in the dataset through our unsupervised analysis. These variables capture important demographic, financial, and behavioral aspects that are critical for understanding the credit loan approval process.

3.2 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to simplify the dataset while retaining the most important information. By transforming the data into a set of orthogonal components, PCA helps us identify patterns, reduce dimensionality, and focus on the most significant features without losing substantial variance. Below, we summarize the key findings of this analysis.

3.2.1 Variance Explained by Principal Components

Each principal component (PC) was evaluated based on the amount of variance it explains. Table 1 presents the standard deviation, proportion of variance, and cumulative variance explained by each component:

Component	Standard Deviation	Proportion of Variance	Cumulative Proportion
PC1	1.5950	0.2827	0.2827
PC2	1.1243	0.1404	0.4231
PC3	1.0061	0.1125	0.5356
PC4	0.9810	0.1069	0.6425
PC5	0.9416	0.0985	0.7410
PC6	0.8699	0.0841	0.8251
PC7	0.7964	0.0705	0.8956
PC8	0.7417	0.0611	0.9567
PC9	0.6242	0.0433	1.0000

Table 1: Variance explained by each principal component.

From the table, it is evident that the first three components (PC1, PC2, and PC3) explain a combined 53.56% of the total variance in the dataset. This indicates that by focusing on just these components, we can capture more than half of the variability in the data, significantly simplifying the dataset without losing key information.

3.2.2 Variance Explained by Principal Components

To understand how much variance is captured by each principal component, we visualized both the cumulative variance explained and the individual contribution of each component. The cumulative variance plot (Figure 3) shows how the inclusion of more components gradually increases the total variance explained by the model. While the first few components contribute significantly to the overall variance, the cumulative increase continues as more components are added, making it challenging to identify a clear cut-off based solely on this plot.

To further explore the importance of each principal component, we also generated a scree plot (Figure 4), which displays the percentage of variance explained by each component in descending order. The first component explains 28.3% of the variance, followed by the second with 14% and the third with

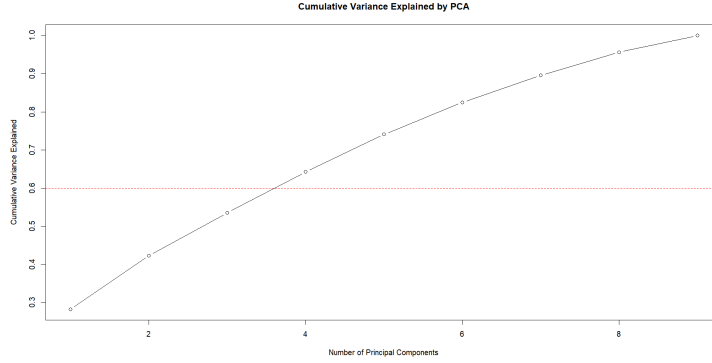


Figure 3: Cumulative variance explained by principal components.

11.2%. After the third component, the contribution of subsequent components decreases gradually. Although the first few components capture the majority of the variance, this gradual decrease suggests that more components contribute meaningfully to the structure of the data. Deciding the optimal number of components to retain will require considering both the cumulative variance explained and the interpretability of the components.

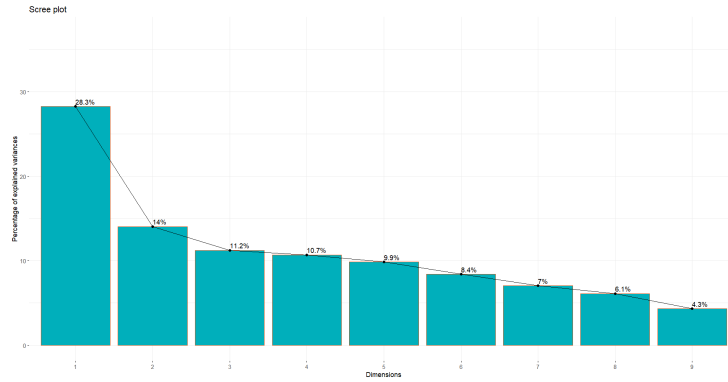


Figure 4: Scree plot of explained variance for principal components.

3.2.3 Selection of Principal Components Based on Eigenvalues

The selection of principal components was performed using the **Kaiser Criterion**, which recommends retaining components with eigenvalues greater than 1. According to this criterion, components with eigenvalues above this threshold account for a substantial portion of the variance in the dataset and are deemed significant for further analysis.

Based on the Kaiser Criterion, the following components were selected:

- **Selected Components:** PC1, PC2, and PC3

These three components have eigenvalues greater than 1 and collectively explain the majority of the variance in the data. The exact eigenvalues for each component are as follows:

- **PC1:** 2.5441
- **PC2:** 1.2640
- **PC3:** 1.0122

Together, these components account for a significant portion of the total variance, confirming their importance in summarizing the dataset. Components with eigenvalues less than 1, such as PC4 (0.9624) and others, explain relatively less variance and were therefore not retained for further analysis.

3.2.4 PCA Biplot

A PCA biplot was generated to visualize the relationships between the observations (individuals) and the variables in the reduced PCA space. The first two principal components (PC1 and PC2), which explain a combined 42.3% of the variance in the dataset, are used to project the data. Observations are colored by the *Approved* status (with blue representing *Rejected* and orange representing *Approved*), and ellipses are drawn around these groups to highlight their clustering.

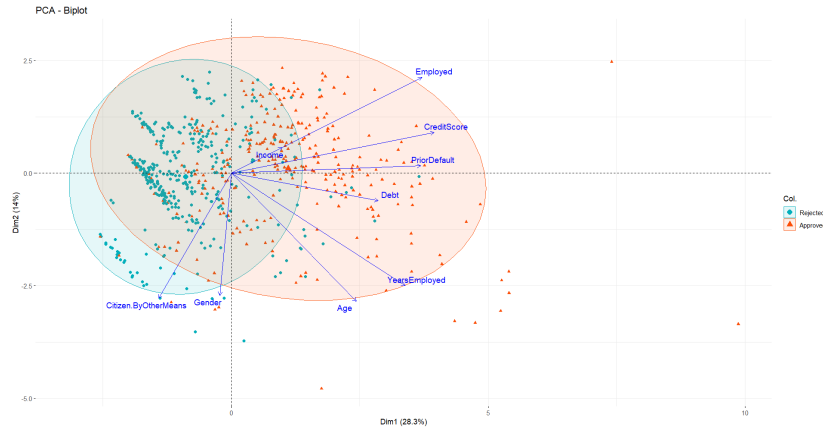


Figure 5: PCA Biplot of Individuals and Variables

Interpretation The biplot above offers insight into how the observations are distributed across PC1 and PC2, as well as how different variables contribute to these components. Key observations from the biplot include:

- **Clustering by Approval Status:** There is a discernible distinction between *Approved* (orange triangles) and *Rejected* (blue circles) applications. The *Approved* group is primarily clustered towards the positive end of PC1 and PC2, whereas the *Rejected* applications are more dispersed, often located on the negative side of these components.
- **Contribution of Variables:**
 - Variables such as *Employed*, *CreditScore*, *PriorDefault*, and *Debt* contribute positively to PC1 and PC2, suggesting that higher values in these variables are associated with *Approved* applications.
 - On the other hand, variables like *Gender*, *Citizen.ByOtherMeans*, and *Age* are associated with the negative sides of PC1 and PC2, indicating that lower values for these variables are linked to *Rejected* applications.
- **Group Separation:** The ellipses around the *Approved* and *Rejected* groups represent confidence intervals, indicating that while the groups can be somewhat separated in the PCA space, there is still noticeable overlap between them. This implies that the first two principal components capture some, but not all, of the variance distinguishing these two groups.
- **Importance of PC1 and PC2:** PC1 accounts for 28.3% of the variance, while PC2 explains 14%. Together, these components capture 42.3% of the total variance, indicating that they provide meaningful insight into the dataset, although higher-order components still hold additional information.

In summary, the PCA biplot offers valuable visualization of the relationships between variables and observations. Variables such as *Employed*, *CreditScore*, and *PriorDefault* are critical in differentiating *Approved* from *Rejected* applications. Meanwhile, variables such as *Gender* and *Citizen.ByOtherMeans* appear more aligned with rejected applications.

3.2.5 Contribution of Variables and PCA Loadings

The contribution of variables and their PCA loadings are essential to understanding how each variable influences the principal components. The **contribution** of a variable indicates the percentage of variance that the variable explains in a particular principal component, while the **loadings** describe the direction and strength of the relationship between the variables and the components.

The table (Table ??) shows the loadings of the first three principal components (PC1, PC2, and PC3). These loadings indicate how much each variable contributes to the corresponding principal component.

Variable	PC1	PC2	PC3
Gender	-0.026	-0.456	0.067
Age	0.287	-0.478	-0.053
Debt	0.338	-0.103	-0.354
YearsEmployed	0.400	-0.420	0.004
PriorDefault	0.436	0.027	0.114
Employed	0.439	0.357	0.246
CreditScore	0.467	0.151	0.221
Income	0.117	0.094	-0.848
Citizen.ByOtherMeans	-0.167	-0.466	0.159

Table 2: PCA Loadings for PC1, PC2, and PC3

Contribution and Loadings for PC1 As shown in the table, variables such as *Employed*, *PriorDefault*, and *YearsEmployed* have the highest loadings in PC1, suggesting that they are the most influential in explaining the variance along this component.

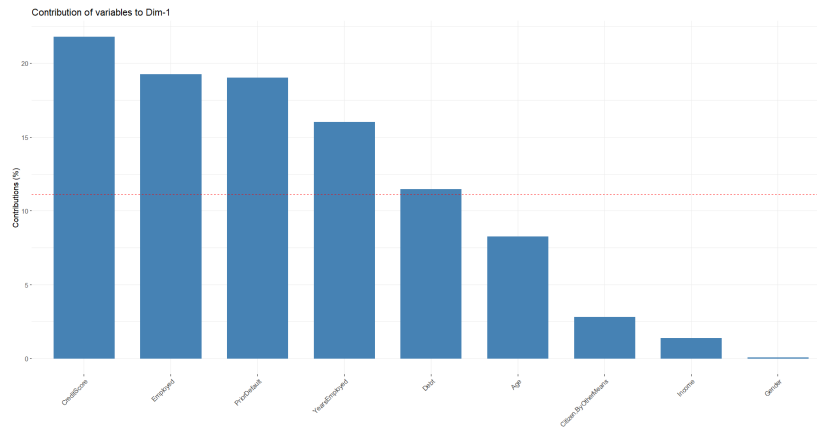


Figure 6: Top 10 Contributions to PC1

These contributions suggest that PC1 captures a dimension of financial stability and employment status, where variables such as *Employed* and *CreditScore* have strong positive loadings, indicating that higher values of these variables correspond to higher PC1 scores.

Contribution and Loadings for PC2 For PC2, the variables with the highest contributions (Figure 7) are:

- **Age** (22.1%)
- **Citizen.ByOtherMeans** (20.6%)

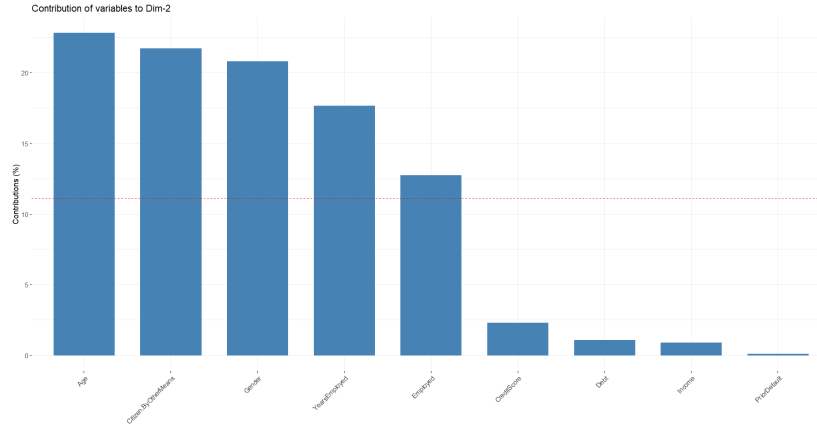


Figure 7: Top 10 Contributions to PC2

- **Gender** (19.6%)
- **YearsEmployed** (17.8%)
- **Employed** (14.5%)

The loadings in Table ?? show that both *Age* and *Gender* have negative loadings (-0.478 and -0.456, respectively), indicating that lower values of these variables are associated with higher scores along PC2. This suggests that PC2 primarily captures a demographic dimension, where younger individuals and certain demographic characteristics, such as gender or citizenship status, contribute more strongly to this component. This distinction highlights demographic variability that may be influential in differentiating credit approval outcomes.

Contribution and Loadings for PC3 For PC3, *Income* is the dominant contributor, accounting for 67.8% of the total contribution (Figure 8). The loadings reflect this, with *Income* having a prominent positive loading (0.246), suggesting that PC3 primarily captures variability related to income levels. Additionally, *Debt* and *Employed* also contribute to PC3, though to a lesser extent. Notably, *Debt* has a negative loading (-0.354), implying that higher debt is associated with lower scores along PC3. This indicates that PC3 may reflect a

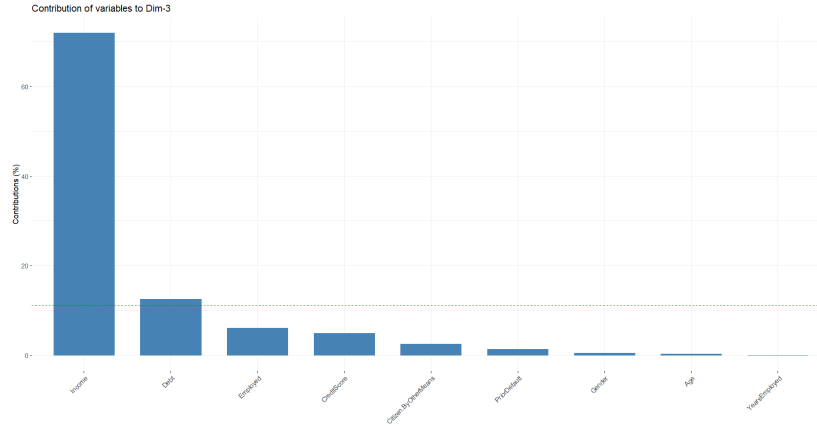


Figure 8: Top 10 Contributions to PC3

3.3 Interpretation of PCA Scores

After analyzing the loadings, which show how the original variables contribute to each principal component, we can now interpret the **PCA scores** to understand how individual observations (or data points) are positioned in the space defined by the first three principal components (PC1, PC2, and PC3). Each score represents the value of an observation along a principal component, allowing us to explore the relationship between individuals and the newly formed axes.

The following table (Table 3) shows the PCA scores for the first six observations in the dataset, focusing on the first three principal components:

Observation	PC1	PC2	PC3
1	0.3424	0.4633	0.8322
2	2.0839	0.1810	0.3842
3	-0.6536	0.9297	0.0038
4	1.0572	0.3629	0.9206
5	-1.0683	-1.7252	0.5153
6	-0.1882	-0.5783	0.0009

Table 3: PCA Scores for the First Six Observations on PC1, PC2, and PC3

Scores on PC1 PC1, as previously discussed, captures a dimension of *financial stability and employment status*, with high contributions from variables such as *Employed*, *PriorDefault*, *CreditScore*, and *Debt*. Observations with high scores on PC1 are likely to have higher values for these variables, indicating greater financial stability or better creditworthiness.

For example, observation 2 has a high score on PC1 (2.08), suggesting that this individual has high values for *CreditScore*, *Employed*, and *PriorDefault*,

reflecting a higher degree of financial stability. On the other hand, observation 5 has a low score on PC1 (-1.07), indicating weaker financial stability, potentially driven by lower employment status or higher debt.

Scores on PC2 PC2 primarily captures a dimension of *demographic variation*, with key contributions from variables such as *Age*, *Citizen.ByOtherMeans*, *Gender*, and *YearsEmployed*. Observations with high scores on PC2 are likely to represent younger individuals or specific demographic categories.

For instance, observation 3 has a high score on PC2 (0.93), which, combined with the negative loading for *Age*, suggests that this observation likely corresponds to a younger individual with lower years of employment. Conversely, observation 5 has a negative score on PC2 (-1.72), potentially representing an older individual with higher years of employment and stability.

Scores on PC3 PC3 is primarily influenced by *Income*, with other contributing variables being *Debt* and *Employed*. Observations with high scores on PC3 are associated with higher incomes, while those with lower scores might have lower incomes or higher debts.

For example, observation 4 has a high score on PC3 (0.92), indicating that this individual likely has a higher income, given the strong positive loading of *Income* on PC3. In contrast, observation 3 has a score near zero on PC3 (0.00), indicating that income may not be a significant differentiating factor for this individual.

3.3.1 Conclusion

Principal Component Analysis (PCA) proved to be an effective technique for reducing the dimensionality of the dataset, allowing us to capture the underlying variance while retaining the most important features. By focusing on the first three principal components (PC1, PC2, and PC3), we were able to explain a significant portion of the variance in the data. PC1, which primarily reflects financial stability and employment status, is driven by variables like *Employed*, *PriorDefault*, and *CreditScore*. PC2, on the other hand, captures demographic variation, where *Age*, *Gender*, and *Citizen.ByOtherMeans* play a major role. Finally, PC3 is largely influenced by *Income*, which highlights economic standing as a key factor.

By examining both the variable loadings and individual PCA scores, we gained valuable insights into how variables and observations are distributed across these components. While the first three principal components explain a significant amount of the dataset's variance, further components also contribute meaningfully, and a balance between explained variance and interpretability guided our focus on the first three.

3.4 Hierarchical Clustering Analysis

To handle the mixed data types in our dataset (both categorical and continuous variables), we used **Gower's distance** for the dissimilarity matrix. Gower's distance is well-suited for mixed data because it standardizes the contribution of each variable type, ensuring that both categorical and numerical variables are appropriately scaled and considered in the distance calculations.

For the hierarchical clustering, we applied the **Ward.D2 method**. This method minimizes the total within-cluster variance and ensures that each step of the clustering process merges the pair of clusters that result in the least increase in the overall variance.

3.4.1 Hierarchical Clustering Dendrogram (Without Predefined Clusters)

Figure 9 presents the dendrogram generated through hierarchical clustering. This dendrogram provides a visual representation of the hierarchical structure of the data, without any predefined number of clusters. Each branch represents the merging of clusters based on similarity, allowing us to see how clusters form at different levels of dissimilarity.

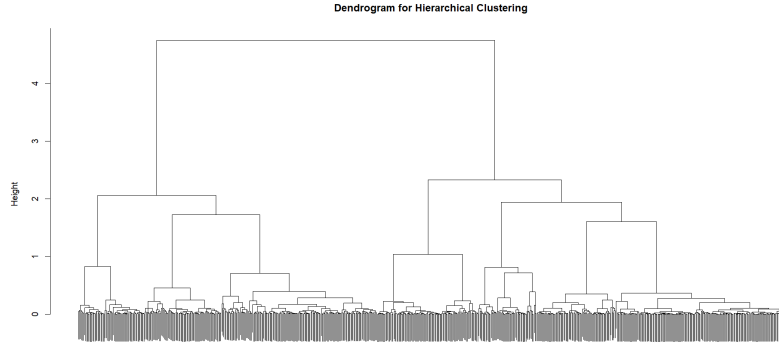


Figure 9: Dendrogram for Hierarchical Clustering (Without Predefined Clusters)

The dendrogram reveals the hierarchical structure of the data, where data points that merge at lower heights are more similar to each other compared to those merging at higher levels. This structure shows us not only the clusters but also how similar or different the clusters are from one another. As we move up the dendrogram, the merging of branches at various heights reflects the degree of dissimilarity, with larger merges indicating more distinct groupings.

We can also use the dendrogram to determine potential cut points for clustering. By visually inspecting where larger gaps in branch heights occur, we

can identify natural divisions within the data. Cutting the dendrogram at these points will define clusters, with each branch below the cut representing a separate group. For instance, significant changes in branch height suggest that these might be appropriate places to cut if we aim to capture major divisions within the data.

This dendrogram further illustrates the process of cluster merging, which begins with each data point as its own individual cluster. As the clustering algorithm progresses, points that are most similar are merged first, and this continues until all data points eventually combine into one single cluster at the top. This process, based on Gower’s distance, highlights the nested nature of the hierarchical clustering approach and provides insight into how clusters evolve as the similarity threshold is relaxed.

Finally, the dendrogram helps us observe overall data patterns and grouping tendencies. By examining the branch structure, we can assess whether the data naturally falls into distinct groups or if it shows a more continuous, evenly spread pattern. This insight is crucial, as it informs us about the underlying structure of the data and helps to confirm whether the choice of a certain number of clusters is appropriate. The plain dendrogram, therefore, offers a comprehensive view of the data’s clustering structure, supporting flexible and informed decisions on cluster selection.

3.4.2 Silhouette Method for Optimal Cluster Evaluation

To evaluate the quality of clustering and determine the optimal number of clusters, the silhouette method was applied. The silhouette score measures how well each data point fits within its assigned cluster relative to the nearest neighboring cluster, with values close to 1 indicating well-clustered points, values near 0 indicating overlapping clusters, and negative values indicating potential misclassification.

Several clustering configurations were tested with different numbers of clusters (k), and the silhouette method revealed that $k = 4$ produced the highest average silhouette width. Figure 10 illustrates the silhouette plot for $k = 4$ clusters, where the average silhouette width is 0.41. Clusters 1, 2, and 4 demonstrate relatively higher silhouette widths, indicating well-defined structures, while Cluster 3 contains points with lower silhouette values, suggesting some overlap or boundary cases.

Based on this analysis, $k = 4$ clusters were selected as the optimal configuration. This choice was guided by the elevated silhouette scores, reflecting a balance between well-defined clustering and interpretability, enabling a meaningful exploration of group characteristics in the dataset.

3.4.3 Dendrogram with Cluster Highlighting

Figure 11 displays the dendrogram generated from hierarchical clustering, with data points grouped into $k = 4$ clusters. In this visualization, each cluster is color-coded, making it easier to distinguish the distinct groups that emerge

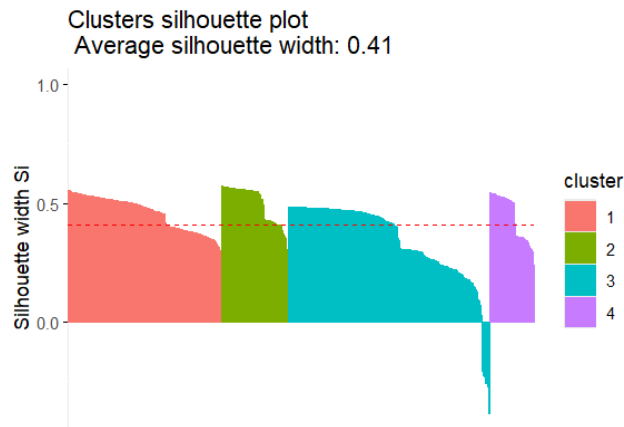


Figure 10: Silhouette Plot for $k = 4$ Clusters

from the dataset. The dendrogram highlights the hierarchical structure of the data, allowing us to visually assess the division of data points into meaningful clusters.

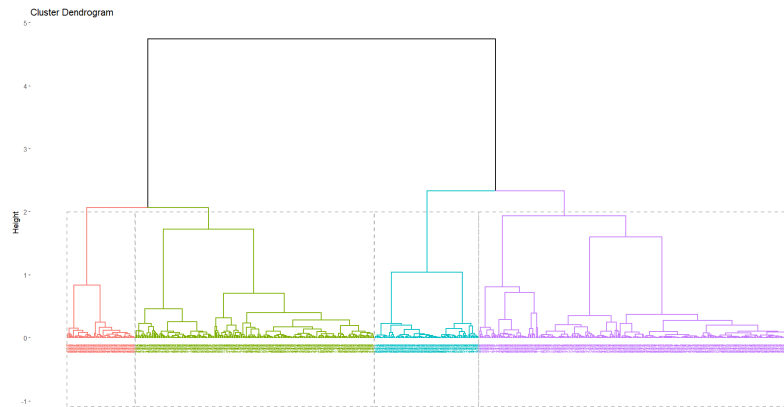


Figure 11: Dendrogram for Hierarchical Clustering with 4 Clusters

Interpretation: The color-coded dendrogram provides a clear visualization of the cluster structure within the data. Each shaded block represents one of the 4 clusters, indicating how the data points are grouped based on their similarity as measured by Gower's distance.

The heights at which clusters merge offer valuable insights. Each merging point represents the level of dissimilarity at which two clusters combine, with

taller nodes indicating more distinct clusters. Large jumps in these heights suggest natural divisions within the data. By focusing on significant merges at larger heights, we identified 4 distinct clusters as the optimal choice, balancing clustering quality and interpretability.

This dendrogram also highlights the hierarchical relationships between clusters, where smaller, similar clusters combine to form larger groups. Some groups merge sooner, while others remain separate until higher levels of dissimilarity are reached. This nested structure reflects meaningful layers within the data, where clusters encapsulate shared attributes among observations.

Cluster Interpretation: To better understand the characteristics of each cluster, we analyze both numeric and categorical variables. The table below summarizes key statistics for each cluster, followed by a detailed interpretation.

Cluster 1: This group consists of individuals with slightly above-average age (Mean = 0.237), higher debt levels (Mean = 0.337) and moderate years of employment (Mean = 0.438). They have relatively high credit scores (Mean = 0.863) with notable variation (SD = 1.34) and slightly above-average incomes (Mean = 0.160). Predominantly male (Gender Mean = 0.670), they have prior defaults (Prior Default Mean = 1) and are employed (Employed Mean = 1). They do not belong to minority groups (Citizen By Other Means Mean = 0).

Feature	Value
Age Mean	0.237
Age SD	1.10
Debt Mean	0.337
Debt SD	1.15
Years Employed Mean	0.438
Years Employed SD	1.28
Credit Score Mean	0.863
Credit Score SD	1.34
Income Mean	0.160
Income SD	1.06
Gender Mode	Male
Prior Default Mode	Yes
Employed Mode	Yes
Citizen By Other Means Mode	No

Table 4: Cluster 1 Summary

Cluster 2: This cluster consists of younger individuals (Age Mean = -0.134) with low debt (Debt Mean = -0.0584) and minimal years of employment (Years Employed Mean = -0.268). Their average credit score is relatively low (Credit Score Mean = -0.494) with no variation (SD = 0), and their income is below average (Income Mean =

-0.0724). Members of this cluster are predominantly female, are less likely to have prior defaults, and are typically unemployed. Additionally, they do not belong to minority groups (Citizen By Other Means Mean = 0).

Feature	Value
Age Mean	-0.134
Age SD	0.957
Debt Mean	-0.0584
Debt SD	0.855
Years Employed Mean	-0.268
Years Employed SD	0.659
Credit Score Mean	-0.494
Credit Score SD	0
Income Mean	-0.0724
Income SD	0.417
Gender Mode	Female
Prior Default Mode	No
Employed Mode	No
Citizen By Other Means Mode	No

Table 5: Cluster 2 Summary

Cluster 3: This cluster is characterized by younger individuals (Age Mean = -0.0563) with relatively low debt levels (Debt Mean = -0.188) and fewer years of employment (Years Employed Mean = -0.171). They have low credit scores (Credit Score Mean = -0.483) with minimal variation (SD = 0.140) and incomes slightly below average (Income Mean = -0.0666). Members of this cluster are predominantly male (Gender Mean = 0.967), unemployed (Employed Mean = 0.010), and unlikely to have prior defaults or belong to minority groups (Citizen By Other Means Mean = 0.191).

Cluster 4: This group is relatively younger (Age Mean = -0.365) with low debt (Debt Mean = -0.221) and minimal years of employment (Years Employed Mean = -0.336). Their credit scores are close to the average (Credit Score Mean = -0.0443), with moderate variation (SD = 0.455), and their income is slightly below average (Income Mean = -0.142). Members of this cluster are predominantly male (Gender Mean = 0.6), are typically employed (Employed Mean = 1), and are unlikely to have prior defaults (Prior Default Mean = 0) or belong to minority groups (Citizen By Other Means Mean = 0).

These cluster interpretations provide a comprehensive view of the distinct groups within the dataset. By understanding the combination of financial, demographic, and employment-related variables, we can tailor strategies to target the specific needs and characteristics of each cluster.

Feature	Value
Age Mean	-0.0563
Age SD	0.909
Debt Mean	-0.188
Debt SD	0.876
Years Employed Mean	-0.171
Years Employed SD	0.789
Credit Score Mean	-0.483
Credit Score SD	0.140
Income Mean	-0.0666
Income SD	1.17
Gender Mode	Male
Prior Default Mode	No
Employed Mode	No
Citizen By Other Means Mode	No

Table 6: Cluster 3 Summary

Feature	Value
Age Mean	-0.365
Age SD	0.946
Debt Mean	-0.221
Debt SD	0.861
Years Employed Mean	-0.336
Years Employed SD	0.620
Credit Score Mean	-0.0443
Credit Score SD	0.455
Income Mean	-0.142
Income SD	0.155
Gender Mode	Male
Prior Default Mode	No
Employed Mode	Yes
Citizen By Other Means Mode	No

Table 7: Cluster 4 Summary

3.5 t-SNE Dimensionality Reduction

To visualize high-dimensional data in a low-dimensional space, we applied **t-distributed Stochastic Neighbor Embedding (t-SNE)** to reduce the dimensionality of our dataset. t-SNE is particularly effective at capturing the local structure of the data and revealing patterns or clusters in two-dimensional or three-dimensional plots.

3.6 t-SNE Dimensionality Reduction

To better understand and visualize the structure of high-dimensional data in a more accessible form, we employed **t-distributed Stochastic Neighbor Embedding (t-SNE)** to reduce the dimensionality of our dataset. t-SNE is a powerful dimensionality reduction technique that is particularly effective at preserving local structures in the data. By projecting the data into a two-dimensional space, it allows us to observe potential clusters or patterns that may not be evident in the original high-dimensional space.

3.6.1 t-SNE Plot with Class Labels

The t-SNE plot in Figure 12 shows a two-dimensional representation of the credit data, reduced through t-SNE. Each point represents an observation, color-coded by approval status: blue for approved and orange for rejected applications. The t-SNE axes, labeled as t-SNE 1 and t-SNE 2, capture complex relationships within the data, emphasizing proximity among similar points.

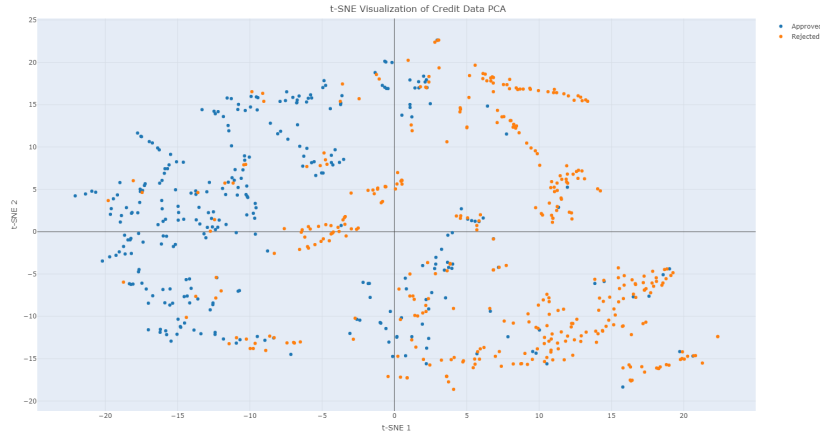


Figure 12: t-SNE Visualization of Credit Data with Approval Status

This plot reveals distinct areas where approved and rejected applications cluster separately, suggesting that certain combinations of features strongly correlate with approval outcomes. Specifically, we observe regions on the left and right where one class is more concentrated, indicating that t-SNE has highlighted some separation between approved and rejected cases. However, in the central part of the plot, blue and orange points intermingle, implying that certain applications share similar characteristics regardless of approval status. These overlaps suggest that classification may be challenging for cases with mixed traits.

Overall, the t-SNE visualization provides valuable insight into the structure of the data, showing clear clusters where the decision is more predictable and areas of overlap where the distinction between approved and rejected applications

is less defined.

3.6.2 t-SNE Plot with K-Means Clustering

To further explore the underlying structure, we applied **K-Means clustering** to the t-SNE-reduced data, specifying 3 clusters. The clustered t-SNE plot (Figure 13) shows the three groups identified by the algorithm.



Figure 13: t-SNE Clustering Visualization (3 Clusters)

Interpretation of Clustering Plot:

- **Cluster Separation:** The K-Means algorithm identifies three distinct clusters in the t-SNE space. These clusters show meaningful separation, but some overlap exists, particularly between clusters 1 and 2.
- **Group Insights:** The separation between clusters indicates that the algorithm was able to distinguish between different groups of data points, though further refinement could improve the clarity of cluster boundaries.

3.6.3 Silhouette Analysis for Cluster Validation

To evaluate the quality of the clusters, we performed a **Silhouette analysis**, which measures the cohesion and separation of clusters. The silhouette score quantifies how well a data point fits within its own cluster versus how well it is separated from other clusters. Higher average silhouette values suggest better-defined clusters.

Silhouette Results:

- Cluster 1: Average silhouette width = 0.389

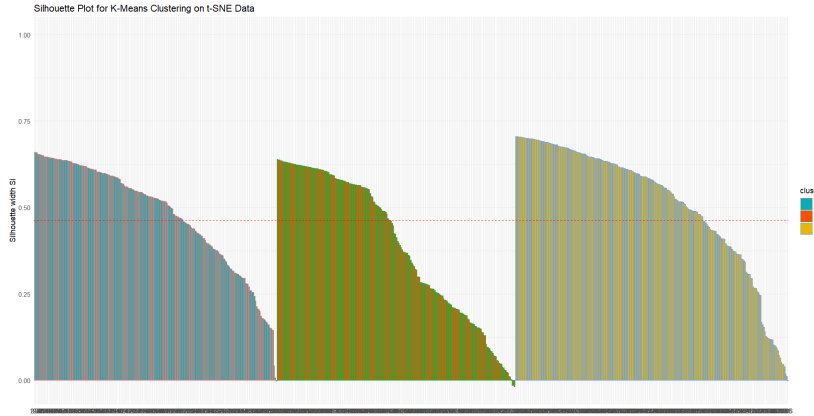


Figure 14: Silhouette Plot for K-Means Clustering on t-SNE Data

- Cluster 2: Average silhouette width = 0.509
- Cluster 3: Average silhouette width = 0.481

Interpretation: Clusters 2 and 3 have relatively high silhouette widths, indicating well-separated clusters with good cohesion. However, cluster 1 has a lower silhouette width (0.389), which suggests that this group may not be as distinct or cohesive as the other two. This could imply that some data points in cluster 1 are not optimally clustered.

We also experimented with a higher number of clusters (e.g., 7 clusters), which produced similar silhouette scores, although the division into 3 clusters provides a simpler and more interpretable result. The decision between 3 and more clusters depends on the level of granularity desired for the analysis.

3.6.4 Cluster 1: Young Individuals with Moderate Debt and Average Credit Score

Feature	Mean	Standard Deviation (SD)
Gender	0.410	0.493
Age	-0.624	0.496
Debt	0.579	1.096
Years Employed	-0.282	0.441
Prior Default	0.502	0.501
Credit Score	-0.113	0.671
Income	-0.110	0.243
Citizen By Other Means	0.037	0.189

Table 8: Cluster 1 Summary: Means and Standard Deviations for Key Features

Cluster 1 primarily consists of younger individuals, as indicated by the negative mean age (Age Mean = -0.624), suggesting they are below the dataset’s average age. These individuals have moderate levels of debt (Debt Mean = 0.579) with relatively high variability (Debt SD = 1.096), indicating a diverse range of financial obligations within the group.

In terms of employment, this cluster shows limited work experience (Years Employed Mean = -0.282) and a lower likelihood of being employed (Employed Mean = 0.438). Their credit scores are slightly below the dataset’s average (Credit Score Mean = -0.113) but exhibit moderate variability (Credit Score SD = 0.671).

Income levels for this group are slightly below average (Income Mean = -0.110), with low variability (Income SD = 0.243). Regarding prior defaults, this group has a moderate likelihood of having defaulted before (Prior Default Mean = 0.502).

Additionally, the cluster includes a relatively balanced gender distribution (Gender Mean = 0.410), and the proportion of individuals categorized as "Citizen By Other Means" is minimal (Mean = 0.037).

This cluster likely represents younger individuals who are early in their financial journeys, with moderate financial obligations, limited work experience, and slightly below-average credit scores.

3.6.5 Cluster 2: Predominantly Male with Low Debt and Poor Credit Scores

Feature	Mean	Standard Deviation (SD)
Gender	0.821	0.384
Age	0.183	0.957
Debt	-0.568	0.347
Years Employed	-0.340	0.429
Prior Default	0.294	0.456
Credit Score	-0.484	0.079
Income	-0.137	0.213
Citizen By Other Means	0.175	0.381

Table 9: Cluster 2 Summary: Means and Standard Deviations for Key Features

Cluster 2 is characterized by predominantly male individuals, as indicated by the high gender mean (Gender Mean = 0.821) and low variability (Gender SD = 0.384). These individuals tend to be slightly older than average (Age Mean = 0.183) with moderate variability in age (Age SD = 0.957).

This cluster has the lowest debt levels among the groups (Debt Mean = -0.568) and relatively low variability (Debt SD = 0.347). However, they have very minimal work experience (Years Employed Mean = -0.340) and lower variability (Years Employed SD = 0.429).

The likelihood of prior defaults is below average (Prior Default Mean = 0.294), suggesting better financial behavior in this regard. However, their credit

scores are significantly below average (Credit Score Mean = -0.484) with very low variability (Credit Score SD = 0.079), indicating consistently poor credit scores across the group.

Income levels are slightly below average (Income Mean = -0.137), with limited variability (Income SD = 0.213). The proportion of individuals categorized as "Citizen By Other Means" is slightly higher than Cluster 1 (Mean = 0.175).

Overall, this cluster represents a group of predominantly male individuals who have low debt and poor credit scores, with minimal work experience and slightly below-average income. This may suggest individuals who are financially stable in terms of debt but are struggling to build credit and increase their earnings.

3.6.6 Cluster 3: Older, Experienced Individuals with High Credit Scores and Income

Feature	Mean	Standard Deviation (SD)
Gender	0.833	0.374
Age	0.404	1.122
Debt	0.079	1.053
Years Employed	0.665	1.441
Prior Default	0.805	0.397
Credit Score	0.663	1.399
Income	0.264	1.709
Citizen By Other Means	0.023	0.149

Table 10: Cluster 3 Summary: Means and Standard Deviations for Key Features

Cluster 3 is composed predominantly of male individuals (Gender Mean = 0.833) with low variability (Gender SD = 0.374). This group tends to be older than average (Age Mean = 0.404) and exhibits a slightly wider age range (Age SD = 1.122).

These individuals have the lowest debt levels across all clusters (Debt Mean = 0.079) with moderate variability (Debt SD = 1.053). They also have the most substantial work experience, as indicated by the high mean years employed (Years Employed Mean = 0.665) and greater variability (Years Employed SD = 1.441).

The likelihood of prior defaults is relatively low (Prior Default Mean = 0.805), suggesting good financial behavior. Additionally, their credit scores are significantly above average (Credit Score Mean = 0.663) with moderate variability (Credit Score SD = 1.399).

Income levels for this group are the highest among all clusters (Income Mean = 0.264) and display notable variability (Income SD = 1.709). Only a small proportion of individuals in this cluster are categorized as "Citizen By Other Means" (Mean = 0.023).

In summary, this cluster represents older, financially stable individuals with

substantial work experience, high credit scores, and above-average income levels. They are likely established in their careers and exhibit strong financial management skills.

3.6.7 Density Plot of t-SNE Clusters

To better understand the spatial distribution of the clusters and the label ('Approved'), we generated a density plot (Figure 15) of the t-SNE clusters. The density plot uses contour lines to show regions of high and low densities for the 'Approved' and 'Rejected' labels.

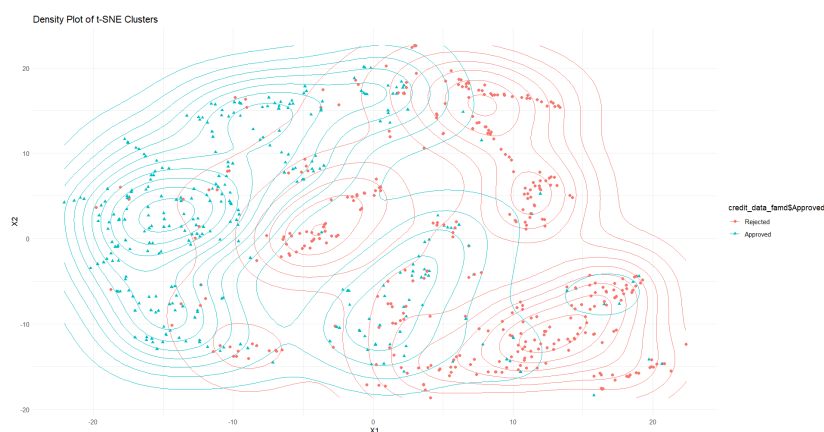


Figure 15: Density Plot of t-SNE Clusters (Approved vs Rejected)

Interpretation: The density plot reveals some separation between the 'Approved' and 'Rejected' classes, particularly in certain regions, suggesting that the t-SNE representation does capture some distinctions between these two categories. The areas of high density for 'Approved' applications tend to cluster more closely in certain regions of the t-SNE space, providing insight into how the features in the dataset relate to loan approval status.

3.7 UMAP Dimensionality Reduction

To further explore the high-dimensional structure of the dataset, we applied **Uniform Manifold Approximation and Projection (UMAP)**. UMAP is an effective non-linear dimensionality reduction technique that excels in preserving both local and global structure of data, making it well-suited for visualizing patterns in two or three dimensions.

3.7.1 UMAP 2D Plot

The UMAP 2D plot (Figure 16) visualizes the dataset reduced to two dimensions. In this plot, each point represents a loan application, and the color indicates whether the application was *approved* (blue) or *rejected* (light red).



Figure 16: UMAP 2D Visualization of Credit Data

Interpretation of UMAP 2D Plot:

- **Cluster Formation:** The UMAP 2D plot reveals certain clusters of approved and rejected applications. Some regions exhibit a clear grouping of either approved or rejected applications, indicating that the UMAP algorithm has captured underlying patterns within the data.
- **Overlap Between Groups:** Despite the visible clustering, there is a significant overlap between the approved and rejected categories. This suggests that the features used for dimensionality reduction may not fully separate the two groups, and further feature engineering or refinement could be necessary.

3.7.2 UMAP 3D Plot

We also extended the UMAP analysis to three dimensions, which is shown in Figure 17. This plot provides a more comprehensive view of the underlying

structure by adding a third dimension, allowing for a better examination of the relationship between approved and rejected applications.

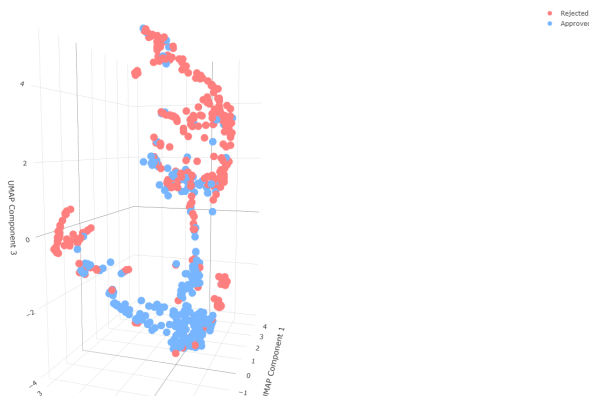


Figure 17: UMAP 3D Visualization of Credit Data

Interpretation of UMAP 3D Plot:

- **Improved Cluster Separation:** The UMAP 3D plot shows clearer cluster separation than the 2D plot, with some distinct groupings of both approved and rejected applications becoming more evident.
- **Remaining Overlap:** However, despite the additional dimension, there is still significant overlap between the two categories in certain areas. This indicates that while UMAP helps to reveal patterns, the decision boundary between approved and rejected applications is not perfectly separable based on the current feature set.

Conclusion: UMAP provides useful visualizations for exploring the high-dimensional structure of the data, but further refinement of the features or alternative methods may be required to achieve more distinct separability between approved and rejected applications.

3.8 Conclusion of Unsupervised Analysis

The unsupervised analysis on this dataset employed several techniques—PCA, hierarchical clustering, t-SNE, and UMAP—to uncover patterns and structures without relying on predefined labels. Each method contributed unique insights, enhancing our understanding of the data’s underlying characteristics.

3.8.1 Principal Component Analysis (PCA)

Through PCA, we selected the first three principal components, which collectively explained 53.56% of the total variance. The loadings indicated the following:

- **PC1** was closely associated with financial variables like *Debt*, *PriorDefault*, and *Employed*, capturing aspects related to financial stability and creditworthiness.
- **PC2** represented demographic factors, with significant contributions from *Age* and *Gender*, indicating socio-demographic influences.
- **PC3** further differentiated individuals based on income and debt-related factors, driven by *Debt* and *Income*.

PCA provided an overview of the key factors driving variance within the dataset, highlighting dimensions of financial stability and demographic influences.

3.8.2 Hierarchical Clustering

To segment the data into distinct groups, hierarchical clustering was performed, with the optimal number of clusters determined using the dendrogram and silhouette analysis. This method suggested 4 clusters, representing natural groupings within the dataset. These clusters highlight subpopulations that differ in terms of financial stability, employment history, and demographic characteristics, providing a nuanced understanding of the data.

3.8.3 t-SNE and UMAP

Both t-SNE and UMAP were applied to project the high-dimensional data into a two-dimensional space, offering complementary perspectives on the dataset:

- **t-SNE:** Focused on preserving local structure, effectively highlighting clusters of approved and rejected applications. While certain regions exhibited clear separations, others showed intermixing, reflecting shared characteristics or transitional states. Cluster-specific patterns revealed:
 - **Cluster 1:** Young individuals with moderate debt and limited work experience, positioned closer to overlapping regions due to shared traits with other groups.
 - **Cluster 2:** Predominantly male with low debt but poor credit scores, forming more isolated clusters with distinct boundaries.
 - **Cluster 3:** Older, financially stable individuals with high income and credit scores, forming compact, well-defined clusters.
- **UMAP:** Provided a balanced view, capturing both local and global relationships. This visualization emphasized:

- Clear separations between clusters, particularly for financially distinct groups like Cluster 3.
- Transitional regions where overlap exists, particularly involving Cluster 1, highlighting shared features between approved and rejected applications.

Together, t-SNE and UMAP confirmed and expanded upon the patterns observed with PCA, illustrating where separation is most distinct and where overlap occurs. These visualizations provided key insights into cluster relationships, reinforcing the value of hierarchical clustering and supporting subsequent analyses.

3.8.4 Summary of Key Findings

In conclusion, the unsupervised analysis revealed that both financial and demographic factors play significant roles in shaping the dataset’s structure. Key themes such as financial stability, employment sector, and demographic attributes consistently emerged across PCA, and clustering analyses, indicating that these dimensions are central to understanding the credit approval process. The identification of four clusters through hierarchical clustering, supported by the Silhouette score, suggests that distinct profiles exist within the dataset, potentially representing various risk categories or target segments. Additionally, the t-SNE and UMAP visualizations offered complementary insights, with t-SNE highlighting fine-grained clusters and UMAP providing a holistic view of the data’s global structure.

Overall, these findings provide a comprehensive understanding of the underlying structure within the dataset, revealing key patterns and relationships among financial, demographic, and employment variables. This unsupervised analysis lays a crucial foundation for the next phase of our study, where we will leverage these insights to inform and enhance supervised learning approaches aimed at modeling and predicting credit approval outcomes. By identifying distinct profiles and segmenting the data into meaningful clusters, we have uncovered the complex interplay of factors influencing approval decisions. These insights not only support targeted strategies for credit assessment but also offer a deeper context for interpreting the results of predictive models in the subsequent analysis.

4 Supervised Analysis

In this section, we will perform a supervised analysis using four different machine learning models: **logistic regression with Elastic Net regularization (glmnet)**, **random forest**, **XGBoost**, and **neural networks**. For each model, we will apply **k-fold cross-validation** to evaluate their performance and ensure generalization. The objective of the analysis is to predict the approval status of credit applications based on the features in the dataset, comparing the performance of these models on accuracy, interpretability, and overall predictive power.

4.1 Logistic Regression with Elastic Net Regularization

In this section, we applied **logistic regression with Elastic Net regularization** using the `glmnet` package. Elastic Net regularization combines both **L1 (Lasso)** and **L2 (Ridge)** penalties, allowing for feature selection and improved generalization. The `alpha` parameter was set to 0.5, balancing between L1 and L2 penalties. We performed 10-fold cross-validation to determine the optimal regularization parameter, λ , which controls the strength of the penalty.

4.1.1 Data Preprocessing

The dataset was prepared as follows:

- **One-hot encoding** was applied to categorical variables to convert them into a binary format.
- The target variable *Approved* was converted into a binary numeric variable: **1 for Approved** and **0 for Rejected**.
- The dataset was **split into training (80%)** and **test (20%)** sets to evaluate model performance.

4.1.2 Cross-Validation and Model Fitting

We used 10-fold cross-validation to find the best regularization parameter λ for the logistic regression model with Elastic Net regularization. The best lambda was identified as:

Best lambda (regularization strength): 0.01565443

4.1.3 Feature Selection

The Elastic Net model shrinks some coefficients to zero, effectively selecting the most important variables. The non-zero coefficient variables from the Elastic Net regularization were identified as follows:

Non-zero coefficient variables: (Intercept) Age YearsEmployed PriorDefault Employed CreditScore DriversLicense ZipCode Income Industry.ConsumerDiscretionary Industry.ConsumerStaples Industry.Financials Industry.Industrials Industry.InformationTechnology Industry.Materials Industry.Real.Estate Industry.Research Ethnicity.Black Ethnicity.Latino

The significant variables with non-zero coefficients are those that contributed to the final model. Below are the coefficients of these variables:

Variable	Coefficient
Intercept	-2.0485
Age	0.0157
YearsEmployed	0.1391
PriorDefault	2.8435
Employed	0.6418
CreditScore	0.3796
DriversLicense	-0.0966
ZipCode	-0.1216
Income	0.7432
Industry.ConsumerDiscretionary	-0.3841
Industry.ConsumerStaples	-0.1707
Industry.Financials	-0.4837
Industry.Industrials	0.0937
Industry.InformationTechnology	0.7490
Industry.Materials	0.1662
Industry.Real.Estate	-0.0135
Industry.Research	-0.0252
Ethnicity.Black	0.0852
Ethnicity.Latino	-1.1996

Table 11: Non-zero Coefficient Variables and Their Values in the Logistic Regression Model

4.1.4 Model Evaluation

The performance of the logistic regression model was evaluated using several metrics, including the **Mean Absolute Error (MAE)**, confusion matrix, and the **Area Under the Curve (AUC)**. The evaluation was performed on the test set.

Mean Absolute Error (MAE) vs Log Lambda Plot In order to assess how the model's performance changes with different levels of regularization, we plotted the **Mean Absolute Error (MAE)** against the logarithm of λ . The plot illustrates that for very low values of λ (on the left side of the plot), the MAE remains relatively constant and low. As λ increases (towards the right

side of the plot), the error starts to rise, indicating over-regularization and a reduction in model performance.

The red vertical line marks the **best lambda** value selected by cross-validation. This λ corresponds to the point where the model achieves the lowest MAE, representing the optimal balance between fitting the data and applying regularization to avoid overfitting.

As observed, the chosen λ value lies on the lower end of the logarithmic scale, highlighting the importance of regularization in improving model generalization while keeping the error minimal.

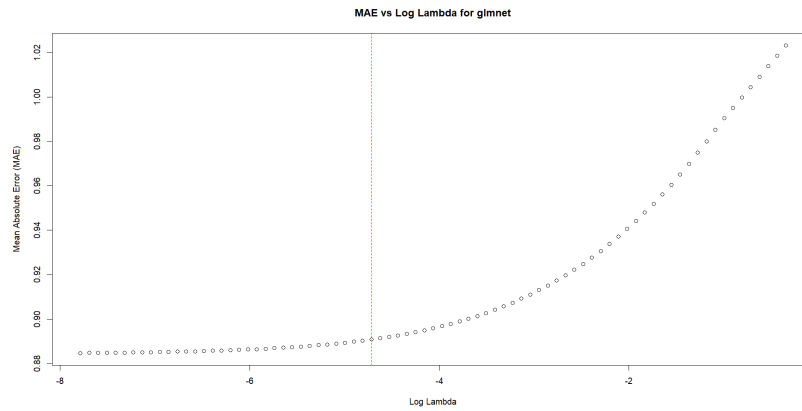


Figure 18: Mean Absolute Error (MAE) vs Log Lambda for Elastic Net Logistic Regression

4.1.5 Model Evaluation on Test Data

To evaluate the model's performance on the test set, we used various metrics, including **Binomial Deviance**, **Misclassification Error**, **Area Under the Curve (AUC)**, **Mean-Squared Error (MSE)**, and **Mean Absolute Error (MAE)**. These metrics provide a comprehensive assessment of the model's classification accuracy, generalization, and overall prediction quality.

The model's performance on the test data is summarized as follows:

- **Binomial Deviance:** 0.7581 — This indicates how well the model fits the data, with lower values suggesting a better fit. Binomial deviance is particularly useful for binary classification tasks like this.
- **Misclassification Error:** 0.1232 — The misclassification error measures the proportion of incorrect classifications. A lower value suggests better model accuracy, and here the error rate is quite low at 12.32%, showing good performance on the test set.

- **AUC (Area Under the ROC Curve):** 0.9339 — The AUC score, which ranges from 0 to 1, provides insight into the model’s ability to distinguish between classes. A high AUC score of 0.9339 indicates that the model has strong discriminatory power, effectively separating approved and rejected loans.
- **Mean-Squared Error (MSE):** 0.2180 — MSE measures the average of the squares of the errors or deviations between predicted and actual values. A low value, such as 0.2180, indicates that the model’s predictions are close to the true outcomes.
- **Mean Absolute Error (MAE):** 0.5584 — MAE measures the average magnitude of the errors in a set of predictions, without considering their direction. This value provides an indication of the average difference between the predicted and actual loan approvals.

The combination of low error rates and a high AUC score demonstrates that the **Elastic Net Logistic Regression** model performed well on the test data, balancing between fitting the training data and generalizing to new, unseen data. The **Binomial Deviance** and **Misclassification Error** are kept low, while the high AUC score confirms the model’s ability to discriminate effectively between classes.

Confusion Matrix The confusion matrix was generated to evaluate the classification performance of the model on the test data. It compares the predicted labels to the actual labels for "Approved" (1) and "Rejected" (0). The model achieved an overall accuracy of **87.68%**, correctly classifying 68 out of 78 rejected applications and 53 out of 60 approved applications.

Predicted \ True	Rejected (0)	Approved (1)	Total
Rejected (0)	68	7	75
Approved (1)	10	53	63
Total	78	60	138

Table 12: Confusion Matrix for Logistic Regression with Elastic Net Regularization

ROC Curve and AUC The **Receiver Operating Characteristic (ROC) curve** was plotted to visualize the performance of the model at different classification thresholds. The **Area Under the Curve (AUC)** is an important metric that quantifies the model’s ability to distinguish between the "Approved" and "Rejected" classes. A higher AUC value indicates better model performance.

The **AUC** value for the Elastic Net logistic regression model was **0.94**, indicating strong classification performance with a good balance between sensitivity and specificity.

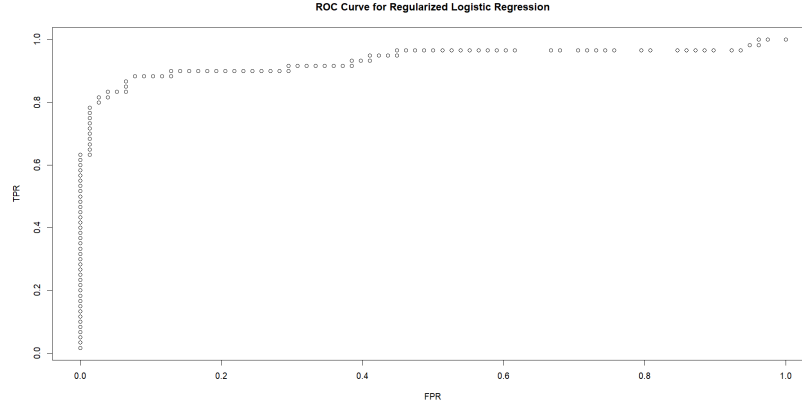


Figure 19: ROC Curve for Elastic Net Logistic Regression

4.1.6 Conclusion

The Elastic Net logistic regression model successfully selected significant features while regularizing others, improving model generalization. The cross-validation process was essential in determining the optimal λ value, which balanced the trade-off between bias and variance. The confusion matrix and ROC curve demonstrated the model's strong classification performance, with the AUC indicating that the model effectively distinguishes between approved and rejected applications.

4.2 Random Forest Classification

In this section, we applied the **Random Forest** algorithm to classify the loan approval status. Random Forest is an ensemble method that combines multiple decision trees to improve model accuracy and prevent overfitting. The training process involved hyperparameter tuning for optimal model performance, with **k-fold cross-validation** and a grid search strategy used to find the best values for **mtry** (number of variables randomly sampled as candidates at each split), **ntree** (number of trees), and **maxnodes** (maximum number of nodes in a tree).

4.2.1 Cross-Validation and Hyperparameter Tuning

The Random Forest model was initially trained using 10-fold cross-validation without tuning **mtry**. Subsequently, we performed a grid search to find the optimal values for **maxnodes** and **ntree**, as these parameters significantly influence model performance. The best values were selected based on the **ROC** metric.

The results of the grid search for **maxnodes** and **ntree** were summarized as follows:

- Best **maxnodes**: 15

- Best **n**tree: 1000

The final model was trained with these best parameters.

4.2.2 Model Evaluation

The final Random Forest model was evaluated on the test dataset, yielding the following performance metrics:

- **Accuracy:** 86.13%
- **Precision (Positive Predictive Value):** 92.00%
- **Recall (Sensitivity):** 75.41%
- **Specificity:** 94.74%
- **F1-Score:** 82.88%

Confusion Matrix The confusion matrix, shown in Table 13, evaluates the classification performance of the Random Forest model on the test dataset, comparing predicted labels to the actual labels for "Approved" and "Rejected" applications. The model correctly identified most approved and rejected applications, though 15 approved applications were misclassified as rejected.

Predicted \ True	Approved (1)	Rejected (0)	Total
Approved (1)	46	4	50
Rejected (0)	15	72	87
Total	61	76	137

Table 13: Confusion Matrix for Random Forest Model

Variable Importance One of the significant advantages of using Random Forest is its ability to estimate the importance of different features in the model. In this analysis, we identified the top 20 most important variables that contributed to predicting the loan approval status.

Table 14 lists the most important variables ranked by their relative importance, with *PriorDefault* being the most influential, followed by *CreditScore*, *Employed*, and *YearsEmployed*. These results suggest that variables related to financial history and employment status play a crucial role in determining loan approval outcomes.

The variable importance plot visually confirms that features related to *PriorDefault*, *CreditScore*, and employment status were among the most critical in predicting whether an application was approved or rejected.

Variable	Importance
PriorDefault	100.000
CreditScore	68.974
Employed	62.424
YearsEmployed	57.003
Income	54.939
Ethnicity.Latino	49.107
Debt	41.854
Industry.InformationTechnology	40.635
ZipCode	36.865
Industry.Real.Estate	34.453
Industry.ConsumerDiscretionary	30.891
Industry.Materials	30.829
Ethnicity.Black	25.699
Citizen.ByOtherMeans	22.753
Age	22.115
Industry.Financials	16.167
Ethnicity.Other	13.774
Ethnicity.Asian	13.404
Gender	11.354
Industry.CommunicationServices	9.825

Table 14: Top 20 Most Important Variables in the Random Forest Model

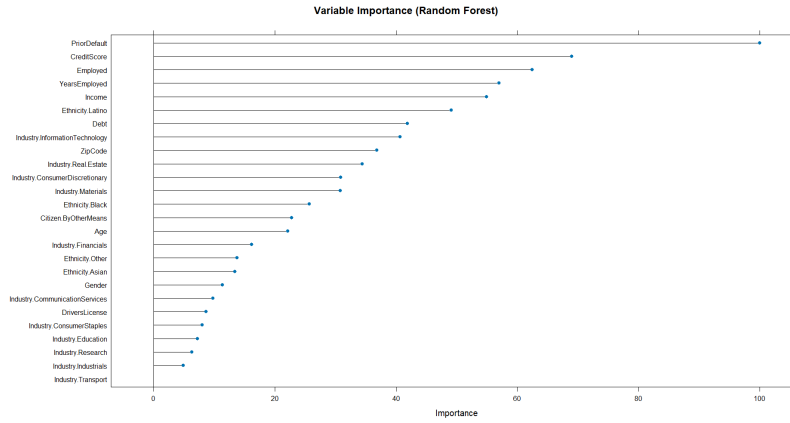


Figure 20: Variable Importance Plot for Random Forest Model

4.3 XGBoost with Cross-Validation

We applied the **XGBoost** algorithm, known for its efficiency and high performance, especially in structured data scenarios. XGBoost uses an ensemble of decision trees to boost classification accuracy. In this analysis, we used **k-fold**

cross-validation to tune hyperparameters and optimize the model's performance.

4.3.1 Data Preprocessing

The dataset was prepared as follows:

- **One-hot encoding** was applied to categorical variables to transform them into numerical features.
- The target variable, *Approved*, was converted into a binary numeric variable: **1 for Approved** and **0 for Rejected**.
- The dataset was **split into training (80%)** and **test (20%)** sets to evaluate the model's performance on unseen data.

4.3.2 Initial Model Training

We first trained a **default XGBoost model** using 100 boosting rounds without any hyperparameter tuning. The default model achieved an accuracy of **86.96%** on the test set, demonstrating the algorithm's strong performance even without parameter optimization.

4.3.3 Hyperparameter Tuning

To further improve the model, we conducted a grid search to tune several hyperparameters:

- **Number of boosting rounds** (`nrounds`): ranging from 100 to 300.
- **Learning rate** (`eta`): 0.025, 0.05, 0.1, 0.3.
- **Maximum tree depth** (`max_depth`): 4, 5, 6.
- **Minimum loss reduction** (`gamma`): 0, 1, 2.
- **Column subsample ratio** (`colsample_bytree`): 0.5, 0.75, 1.0.
- **Minimum sum of instance weight** (`min_child_weight`): 1, 3, 5.

After performing **4-fold cross-validation**, the best hyperparameters were selected based on maximizing the **ROC AUC** score.

Final Model Performance The final XGBoost model was evaluated using a confusion matrix on the test data. The confusion matrix provides detailed insights into the classification performance of the model. The results are as follows:

Prediction	Approved (1)	Rejected (0)
Approved (1)	45	3
Rejected (0)	15	75

Table 15: Confusion Matrix for XGBoost Model on Test Data

Model Performance Metrics The following performance metrics were obtained based on the confusion matrix:

- **Accuracy:** 86.96%
- **Precision:** 83.33% (Positive Predictive Value)
- **Recall:** 96.15% (Sensitivity)
- **Specificity:** 75.00%
- **F1-Score:** 89.29%

Detailed Metrics:

- **Accuracy:** The model achieved an accuracy of 86.96%, meaning that 86.96% of the predictions were correct.
- **Kappa:** A Kappa value of 0.7283 indicates substantial agreement between the predictions and actual values, adjusting for the agreement expected by chance.
- **Sensitivity (Recall):** The sensitivity, or true positive rate, is 96.15%, indicating that the model correctly identified 96.15% of the "Approved" cases.
- **Specificity:** The specificity, or true negative rate, is 75.00%, meaning the model correctly identified 75.00% of the "Rejected" cases.
- **Positive Predictive Value (Precision):** The precision of 83.33% reflects the proportion of true "Approved" cases among all cases predicted as "Approved."
- **F1-Score:** The F1-Score of 89.29% balances both precision and recall, providing an overall measure of model performance for handling the "Approved" class.

These results show that the XGBoost model performs well in classifying credit card applications, with strong recall and balanced accuracy across both approved and rejected cases.

Feature Importance The significance of the features used in the XGBoost model was evaluated using a feature importance plot. This plot showcases the relative contribution of each feature to the final model. The feature **PriorDefault** was identified as the most important predictor, followed by **Income**, **ZipCode**, and **CreditScore**, as illustrated in Figure 21. These features play a significant role in determining the loan approval status.

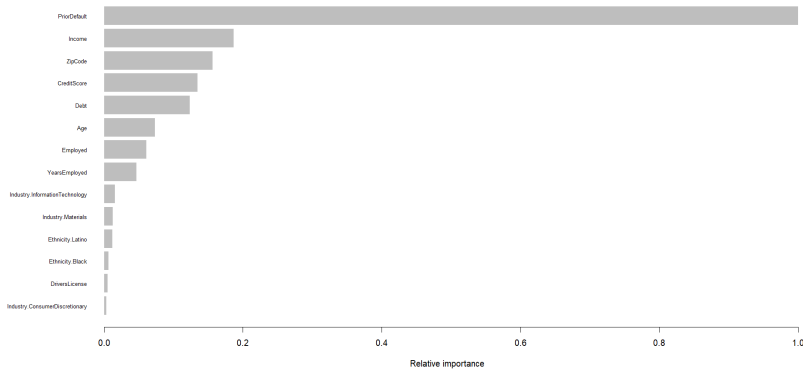


Figure 21: Feature Importance for XGBoost Model

Interpretation of Feature Importance:

- **PriorDefault:** This is the most influential feature in the model, underscoring that a previous default is a key determinant for predicting credit approval.
- **Income:** The second most important feature, highlighting that an applicant's income level is a strong indicator of approval likelihood.
- **ZipCode, CreditScore, Debt:** These features are crucial for capturing geographical, financial, and credit history factors that influence the approval decision.
- **Other Features:** Features like **Employed**, **Age**, and **YearsEmployed** contribute but are less influential compared to the top factors.

This analysis highlights that the XGBoost model efficiently identifies and weighs the most relevant features that affect loan approval, emphasizing the importance of credit history, financial standing, and employment status in the decision-making process.

4.4 Neural Network

A **neural network model** was trained using k-fold cross-validation (10 folds) to predict credit approval status. The **nnet** method was applied, and the dataset was preprocessed by centering and scaling the features to improve convergence.

4.4.1 Cross-Validation Results

The cross-validation process explored different values of two tuning parameters:

- **Size:** The number of neurons in the hidden layer.
- **Decay:** The weight decay, a regularization parameter used to prevent overfitting.

The results of cross-validation are summarized in Table 16. The neural network with **1 hidden neuron** and **decay = 0.1** achieved the highest accuracy, which was selected as the final model.

Size	Decay	Accuracy	Kappa
1	0.0	0.8353	0.6691
1	0.0001	0.8335	0.6658
1	0.001	0.8370	0.6722
1	0.01	0.8297	0.6563
1	0.1	0.8424	0.6828

Table 16: Cross-Validation Results for Neural Network (Top 5 Models)

4.4.2 Confusion Matrix and Model Performance

The neural network model was evaluated on the test dataset, and the performance is summarized in the confusion matrix below. This matrix compares the predicted labels to the true labels and provides detailed classification metrics.

Prediction	Approved (1)	Rejected (0)
Approved (1)	72	10
Rejected (0)	6	50

Table 17: Confusion Matrix for Neural Network Model on Test Data

Model Performance Metrics

- **Accuracy:** 0.8841
- **Precision:** 0.8780
- **Recall:** 0.9231

- **Specificity:** 0.8333
- **F1-Score:** 0.8929
- **Kappa:** 0.7623

Balanced Accuracy and Other Key Metrics The model achieved a **balanced accuracy** of 0.8782, indicating good performance in both correctly identifying positive ("Approved") and negative ("Rejected") classes. The model's sensitivity (recall) is high at 0.9231, meaning it effectively identifies the positive class. The precision score of 0.8780 suggests that the majority of predictions for the "Approved" class are correct.

Other performance metrics such as the **negative predictive value** (0.8929) and **specificity** (0.8333) reinforce the model's ability to generalize well across different cases in the dataset.

The McNemar's test P-value (0.4533) indicates no statistically significant difference between the model's errors on the two classes, further supporting the balanced nature of the predictions.

4.5 Conclusion of Supervised Analysis

In this supervised analysis, we applied a variety of machine learning algorithms—logistic regression with Elastic Net regularization, random forest, XGBoost, and neural networks—to model credit approval outcomes. Each model was fine-tuned using k-fold cross-validation to ensure optimal performance and avoid overfitting.

The logistic regression with Elastic Net regularization selected a subset of significant features, such as Prior Default, Credit Score, and Income, demonstrating strong predictive performance with an AUC of 0.9339. However, the random forest model excelled in variable importance analysis, highlighting Prior Default, Income, and Credit Score as the most influential variables, with an overall accuracy of 86.13.

The XGBoost model, optimized with hyperparameter tuning, achieved a similar performance with an accuracy of 86.96%, while efficiently capturing complex relationships between variables. Finally, the neural network model, despite its simplicity with a single hidden layer, performed well with an accuracy of 88.41%, indicating that even with minimal complexity, neural networks can be competitive in classification tasks.

Each algorithm provided valuable insights and demonstrated strong classification capabilities, but the choice of model depends on the specific goals of the analysis. For interpretability and feature selection, logistic regression and random forest were highly effective, whereas for model performance, XGBoost and neural networks offered competitive results.

This comprehensive analysis not only highlighted the predictive power of different models but also the importance of feature importance and hyperparameter tuning in achieving robust and generalizable models for credit approval prediction.

5 General Conclusion of Supervised and Unsupervised Analysis

This project explored both **supervised** and **unsupervised** learning techniques to predict credit approval and uncover hidden patterns within the dataset. Each method provided unique insights and value to the overall analysis.

5.1 Unsupervised Analysis

The unsupervised analysis employed several techniques to uncover the underlying structure of the credit data, revealing valuable insights into the features that differentiate individuals based on their credit profiles. Starting with **PCA**, we explored the primary sources of variance in the data. PCA identified three main components, which together captured 53.56% of the total variance, emphasizing dimensions related to financial stability, demographic characteristics, and employment history.

Building on these findings, we used **hierarchical clustering** to segment the data, visualizing the results through dendrograms. The Silhouette score method determined an optimal solution of four clusters, each representing a distinct grouping based on shared attributes. This clustering underscored variations in financial stability, demographic profiles, and employment patterns, offering insight into the diversity of credit profiles within the dataset.

To visualize the clusters and confirm their validity, we applied **t-SNE** and **UMAP** for dimensionality reduction, enabling us to project the data into two-dimensional space. Both methods provided complementary perspectives: t-SNE preserved local relationships and highlighted distinct clusters, while UMAP offered a balanced view that captured both local and global structures. Together, these visualizations revealed areas of clear separation and regions of intermixing, suggesting that certain applications share characteristics across approval statuses. This insight is especially valuable for identifying cases where classification may be more complex due to overlapping traits.

The cluster structure was further examined with **k-means clustering** and validated through **silhouette analysis**, which confirmed the cohesion of the four clusters and helped to evaluate the natural separation within the data. These analyses provided a clearer understanding of well-defined subgroups, showing that the dataset consists of diverse profiles and suggesting specific risk categories within both approved and rejected applications.

In conclusion, this unsupervised analysis has effectively identified the latent structure of the credit data, emphasizing the role of financial, demographic, and employment factors in defining distinct subgroups. The insights gained here establish a foundation for feature engineering and will be crucial in guiding the supervised learning phase, where predictive models can be built upon these patterns to better understand and predict credit approval outcomes. This analysis not only informs targeted credit assessment strategies but also offers a comprehensive view of the complex, multidimensional nature of the data.

5.2 Supervised Analysis

The supervised analysis employed several classification algorithms—**logistic regression with Elastic Net regularization**, **random forest**, **XGBoost**, and **neural networks**—to predict credit approval. Each model underwent hyperparameter tuning and cross-validation to optimize performance. The key findings from the supervised analysis include:

- **Logistic regression** with Elastic Net regularization selected significant features such as *Prior Default* and *Credit Score*, achieving an **AUC of 0.9339**. It provided a transparent model that highlighted the most relevant variables.
- **Random forest** emphasized variable importance, showcasing *Prior Default* and *Income* as crucial factors in credit approval prediction. This model demonstrated an **accuracy of 86.13%**.
- **XGBoost**, after extensive hyperparameter tuning, proved highly effective, achieving an **accuracy of 86.96%** and offering both performance and flexibility in handling non-linear relationships between variables.
- The **neural network**, despite being a relatively simple model with only one hidden layer, achieved the highest performance in terms of **accuracy at 88.41%**, highlighting the potential of even basic neural networks for classification tasks in structured datasets.

5.3 Overall Insights

The integration of both unsupervised and supervised learning techniques provided a robust framework for analyzing the dataset, revealing valuable insights into the underlying patterns and predictive factors influencing credit approval. Unsupervised methods, such as clustering and dimensionality reduction, allowed us to uncover natural groupings and explore the latent structure within the data. These techniques proved instrumental in guiding feature selection and enhancing our understanding of the key dimensions—such as financial stability, demographic profiles, and employment characteristics—that differentiate individuals.

On the other hand, supervised learning models demonstrated strong predictive capabilities, offering clarity on the specific factors most strongly associated with credit approval outcomes. By evaluating feature importance and model performance, we gained a deeper understanding of the drivers behind credit decisions, translating data patterns into actionable insights. These models not only confirmed trends observed in the unsupervised analysis but also allowed for a precise assessment of each variable's impact on the prediction outcomes.

Together, these approaches—exploratory and predictive—formed a comprehensive analytical strategy. The unsupervised analysis revealed hidden patterns that enriched our understanding of the dataset's structure, while super-

vised methods provided reliable predictions and interpretable results essential for practical decision-making.